VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

LAB REPORT
on

Machine Learning (23CS6PCMAL)

Submitted by

Dhanush S(1BM23CS089)

in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
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Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Lab work entitled "Machine Learning (23CS6PCMAL)" carried out by **Dhanush S (1BM23CS089),** who is a bonafide student of **B.M.S. College of Engineering.** It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum. The Lab report has been approved as it satisfies the academic requirements in respect of an Machine Learning (23CS6PCMAL) work prescribed for the said degree.

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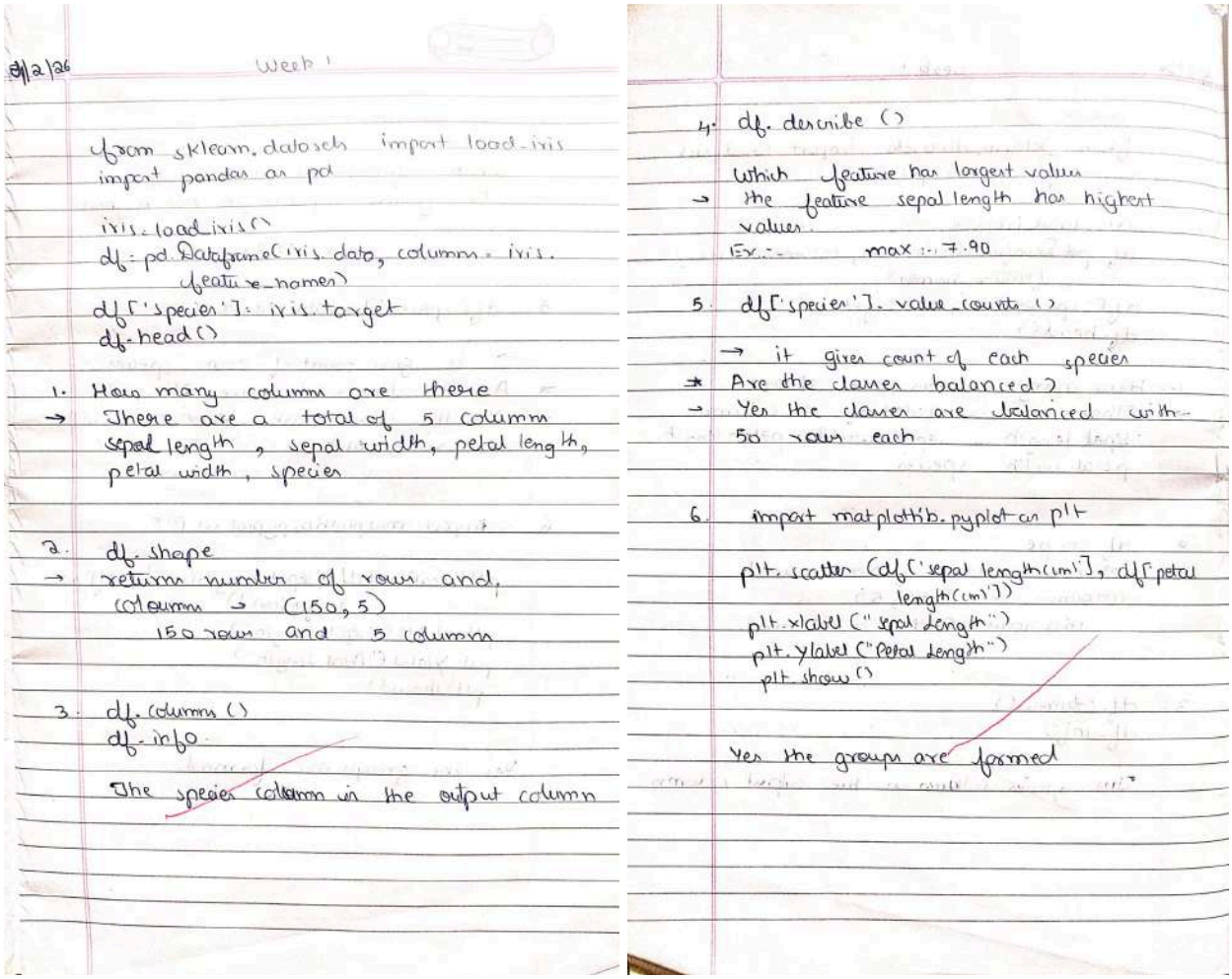
Github Link:
https://github.com/DhanushSGouda/ML_LAB.git

Program 1

Write a python program to import and export data using Pandas library functions

Screenshot



8/2/26 Week 1

from sklearn.datasets import load_iris
import pandas as pd

iris = load_iris()
df = pd.DataFrame(iris.data, columns = iris.feature_names)
df['species'] = iris.target
df.head()

1. How many column are there
→ There are a total of 5 column
sepal length, sepal width, petal length, petal width, species

2. df.shape
→ return number of rows and column → (150, 5)
150 rows and 5 column

3. df.columns()
df.info
The species column in the output column

4. df.describe()
Which feature has largest values
→ the feature sepal length has highest values
Ex:- max :- 7.90

5. df['species'].value_counts()
→ it gives count of each species
* Are the classes balanced?
→ Yes the classes are balanced with 50 rows each

6. import matplotlib.pyplot as plt
plt.scatter(df['sepal length(cm)'], df['petal length(cm)'])
plt.xlabel("sepal length")
plt.ylabel("Petal length")
plt.show()
Yes the groups are formed

7. df[df['species'] == 0].head()
df[df['species'] == 1].head()

Do feature values differ?
Yes the feature values differ the sepal length and petal width of species 1 is more than that of species 0

8. df.corr()

* Which features are correlated?
→
sepal length and petal length :- 0.87
petal width and petal length : 0.96
petal width and sepal length : 0.81

These are positively correlated

9. Sort the data. What happens to species label?

df.sort_values(by='petal length (cm)').head()

When we sort the data by petal length we can see the data sampler with species 0 at the top, thus we can infer that the data sampler with species = 0 have smallest petal length

Code:

```
import pandas as pd
import numpy as np
def process_storm_data():
try:
df = pd.read_csv('cyclone_mocha_data.csv', na_values=[' ', '-', 'N/A'])
#df = pd.read_excel('coastal_erosion_stats.xlsx', sheet_name='Odisha_2024')
print("--- Data Successfully Imported ---")
print(df.head())
except FileNotFoundError:
data = {
'Date': ['12.05.23', '12.05.23', '13.05.23'],
'Time_UTC': ['1800', '2100', '0000'],
'Lat': [15.1, 15.2, 15.4],
'Long': [88.8, 88.9, 89.1],
'Central_Pressure_hPa': [954, 954, 954],
'Wind_Speed_kt': [100, 100, 100],
'Grade': ['ESCS', 'ESCS', 'ESCS']
}
df = pd.DataFrame(data)
print("--- Created Sample DataFrame from Scratch ---")
df['Wind_Speed_kmh'] = df['Wind_Speed_kt'] * 1.852
9
```

```
df.to_csv('processed_storm_data.csv', index=False)
df.to_json('storm_api_output.json', orient='records', indent=4)
# Export to Excel for stakeholder reports (e.g., NDRF or IMD)
# Requires 'openpyxl' library: pip install openpyxl
df.to_excel('storm_summary_report.xlsx', index=False, sheet_name='Summary')
print("\n--- Data Successfully Exported to CSV, JSON, and Excel ---")
if __name__ == "__main__":
process_storm_data()
```

Program 2

Demonstrate various data pre-processing techniques for a given dataset

Screenshot

11/2/26

E to E Machine learning

Step 1: Import the Dataset.

* California housingprice dataset is builtin dataset in sklearn

Step 2: Look at big picture

Here we are using Supervised Learning as the model is being trained on labelled dataset i.e input and output is already known

As house price needs to be calculated - it is Regression

RMSE - Root Mean Square Error
MAE - Mean Absolute Error
R^2 - how much variance is explained

Step 3: Get Data
import datasets

Step 2: Explore Data
→ Do EDA
Ex: - df.describe()
info()

Step 4: Visualize the data.
Map histogram
Correlation Maps
heatMaps

Step 5: Prepare the data

Raw data is messy. Models need clean, structured input, create new features etc

Step 6: Train the Baseline Model

Linear Regression() - Tries to fit using a straight line

Step 7: Random Forest Regressor
Build trees and take averages.

Step 8: Fine Tune the model

Using Hyper parameter Tuning, resulting in better results

Step 9: Calculate the errors to model comparison

	Model	RMSE
0	LR	0.744
1	DT	0.71
2	Random Forest	0.512
3	Tuned RF	0.513

★ Which model performed best?

Random Forest performed well.
As the

★ Why did RM outperform Linear Regression?

As the LR uses a linear relationship it works best on linear data
As the features on which it depends are non-linear

places near coast - more value
more income - more price

Hence Random Forest works well as it builds many decision trees, captures more interactions, averages many trees

★ How did tuning performance increase

In the current scenario the error increased a bit suggesting the default RF settings were more precise

★ What additional data could improve

★ Better location features, (Distance to ocean, city center, neighbourhood quality etc)

→ Economic factors - Interest rates, Infra dev
→ Property features - (Lot size, property tax etc

Code:

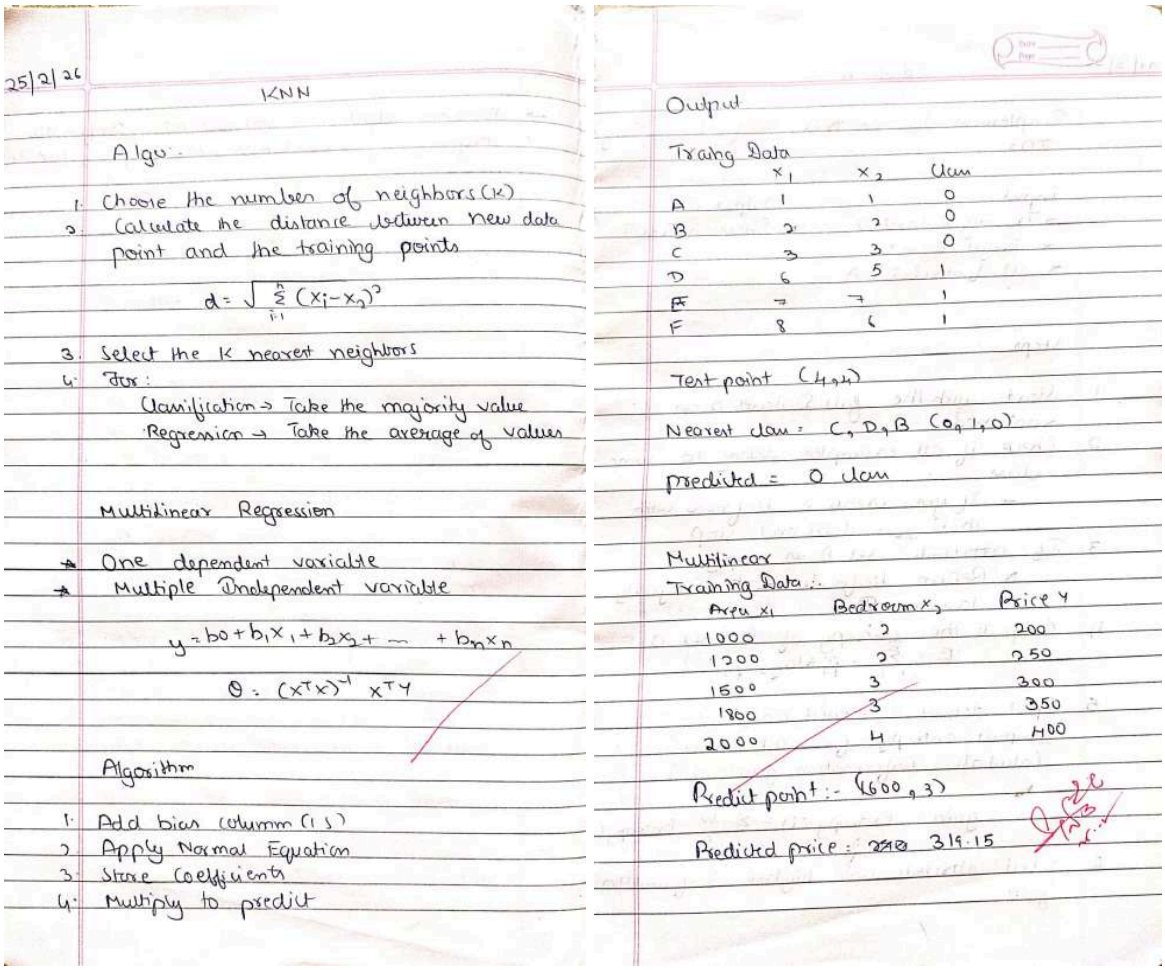
```
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler, MinMaxScaler, LabelEncoder
data = {
'Age': [25, 30, 35, np.nan, 200, 22, 38, 45, 12, np.nan], # Contains NaN and Outlier (200)
'Salary': [50000, 54000, np.nan, 62000, 58000, 52000, 120000, 60000, 48000, 55000],
'Department': ['IT', 'HR', 'IT', 'Sales', 'HR', 'IT', np.nan, 'Sales', 'IT', 'HR'],
'Purchased': ['No', 'Yes', 'No', 'Yes', 'No', 'Yes', 'No', 'No', 'No', 'Yes']
}
df = pd.DataFrame(data)
print("Original Dataset:\n", df)
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Salary'] = df['Salary'].fillna(df['Salary'].mean())
df['Department'] = df['Department'].fillna(df['Department'].mode()[0])
le = LabelEncoder()
```

```
df['Purchased'] = le.fit_transform(df['Purchased'])
df = pd.get_dummies(df, columns=['Department'], drop_first=True)
1
5
Q1 = df['Age'].quantile(0.25)
Q3 = df['Age'].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
df['Age'] = np.where(df['Age'] > upper_bound, upper_bound,
np.where(df['Age'] < lower_bound, lower_bound, df['Age']))
```

Program 3

Implement Linear and Multi-Linear Regression algorithm using appropriate dataset

Screenshot



25/2/26

KNN

Algo:

1. Choose the number of neighbors (K)
2. Calculate the distance between new data point and the training points

$$d = \sqrt{\sum_{i=1}^{n} (x_i - x_2)^2}$$

3. Select the K nearest neighbors
4. For:
   Classification → Take the majority value
   Regression → Take the average of values

Multilinear Regression

* One dependent variable
* Multiple Independent variable

$$y = b_0 + b_1 x_1 + b_2 x_2 + \dots + b_n x_n$$

$$\theta = (X^T X)^{-1} X^T Y$$

Algorithm

1. Add bias column (1s)
2. Apply Normal Equation
3. Store Coefficients
4. Multiply to predict

Output

Traing Data

	x_1	x_2	Class
A	1	1	0
B	2	2	0
C	3	3	0
D	6	5	1
E	7	7	1
F	8	6	1

Test point (4,4)

Nearest class = C, D, B (0, 1, 0)

predicted = 0 class

Multilinear

Training Data:

Area x_1	Bedroom x_2	Price Y
1000	2	200
1200	2	250
1500	3	300
1800	3	350
2000	4	400

Predict point:- (600, 3)

Predicted price: 319.15

Code:

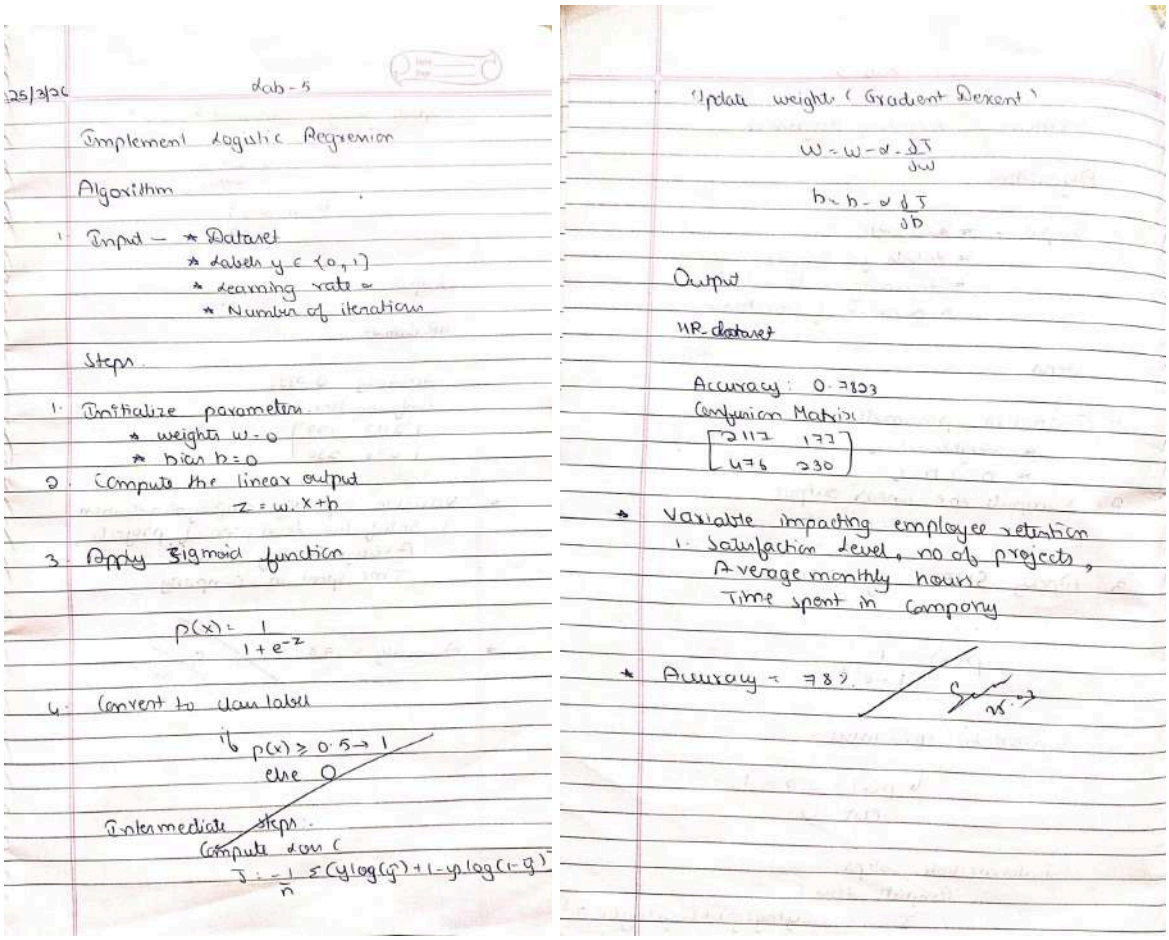
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
df_canada = pd.read_csv('canada_per_capita_income.csv')
X = df_canada[['year']]
y = df_canada['per capita income (US$)']
model = LinearRegression()
model.fit(X, y)
prediction_2020 = model.predict([[2020]])
print(f"Predicted Per Capita Income for 2020: ${prediction_2020[0]:.2f}")
import pandas as pd
from sklearn.linear_model import LinearRegression
df_salary = pd.read_csv('salary.csv')
df_salary['YearsExperience'] =
```

```
df_salary['YearsExperience'].fillna(df_salary['YearsExperience'].median())
X_salary = df_salary[['YearsExperience']]
y_salary = df_salary['Salary']
model_salary = LinearRegression()
model_salary.fit(X_salary, y_salary)
predicted_salary = model_salary.predict([[12]])
print(f"Predicted Salary for 12 years of experience: ${predicted_salary[0]:.2f}")
```

Program 4

Build Logistic Regression Model for a given dataset

Screenshot



25/3/26 Lab - 5

Implement Logistic Regression

Algorithm

1. Input – * Dataset
   * Labels $y \in \{0, 1\}$
   * Learning rate α
   * Number of iterations

Steps

1. Initialize parameters
   * weights $w = 0$
   * bias $b = 0$
2. Compute the linear output
   $z = w \cdot x + b$
3. Apply sigmoid function
   $p(x) = \frac{1}{1+e^{-z}}$
4. Convert to class label
   if $p(x) \geq 0.5 \rightarrow 1$
   else 0

Intermediate steps:
Compute cost C
$J = -\frac{1}{n} \Sigma (y \log(\hat{y}) + (1-y) \log(1-\hat{y}))$

Update weights (Gradient Descent)

$w = w - \alpha \cdot \frac{\partial J}{\partial w}$

$b = b - \alpha \frac{\partial J}{\partial b}$

Output

HR-dataset

Accuracy: 0.7833

Confusion Matrix

$\begin{bmatrix} 2117 & 177 \\ 476 & 230 \end{bmatrix}$

* Variable impacting employee retention
  1. Satisfaction level, no of projects, Average monthly hours, Time spent in company

* Accuracy = 78%

25.3

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
# 1. Load the dataset and Preprocessing
# Note: Ensure the csv files are in your working directory
try:
zoo_df = pd.read_csv('zoo-data.csv')
# Preprocessing:
# Usually, 'animal_name' is a unique identifier and should be dropped
# as it provides no predictive value.
if 'animal_name' in zoo_df.columns:
zoo_df = zoo_df.drop('animal_name', axis=1)
```

```
# Separate features (X) and target (y)
X = zoo_df.drop('class_type', axis=1)
y = zoo_df['class_type']
# Split into Training (80%) and Testing (20%) sets
2
9
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
# 2. Build Logistic Regression Model
# We use multi_class='multinomial' for targets with more than two categories
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
# 3. Measure the accuracy
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {accuracy * 100:.2f}%")
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
# 4. Plot the Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(10, 7))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel('Predicted Class')
plt.ylabel('Actual Class')
plt.title('Confusion Matrix - Zoo Animal Classification')
plt.show()
3
0
except FileNotFoundError:
print("Error: Please ensure 'zoo-data.csv' is in the same folder as this script.")
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
# 1. Load the dataset and Preprocessing
# Note: Ensure the csv files are in your working directory
try:
zoo_df = pd.read_csv('zoo-data.csv')
# Preprocessing:
# Usually, 'animal_name' is a unique identifier and should be dropped
# as it provides no predictive value.
if 'animal_name' in zoo_df.columns:
zoo_df = zoo_df.drop('animal_name', axis=1)
# Separate features (X) and target (y)
3
1
X = zoo_df.drop('class_type', axis=1)
```

```
y = zoo_df['class_type']
# Split into Training (80%) and Testing (20%) sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
# 2. Build Logistic Regression Model
# We use multi_class='multinomial' for targets with more than two categories
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
# 3. Measure the accuracy
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {accuracy * 100:.2f}%")
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
# 4. Plot the Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(10, 7))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel('Predicted Class')
3
2
plt.ylabel('Actual Class')
plt.title('Confusion Matrix - Zoo Animal Classification')
plt.show()
except FileNotFoundError:
print("Error: Please ensure 'zoo-data.csv' is in the same folder as this script.")
```

Program 5

Use an appropriate data set for building the decision tree (ID3) and apply this knowledge to classify a new sample.

Screenshot

11/3/26 Lab - 4

Implement decision tree Classification using ID3

Input
* Training Dataset D
* Target class Y
* Set of attributes A

Output
* Decision Tree T

Steps

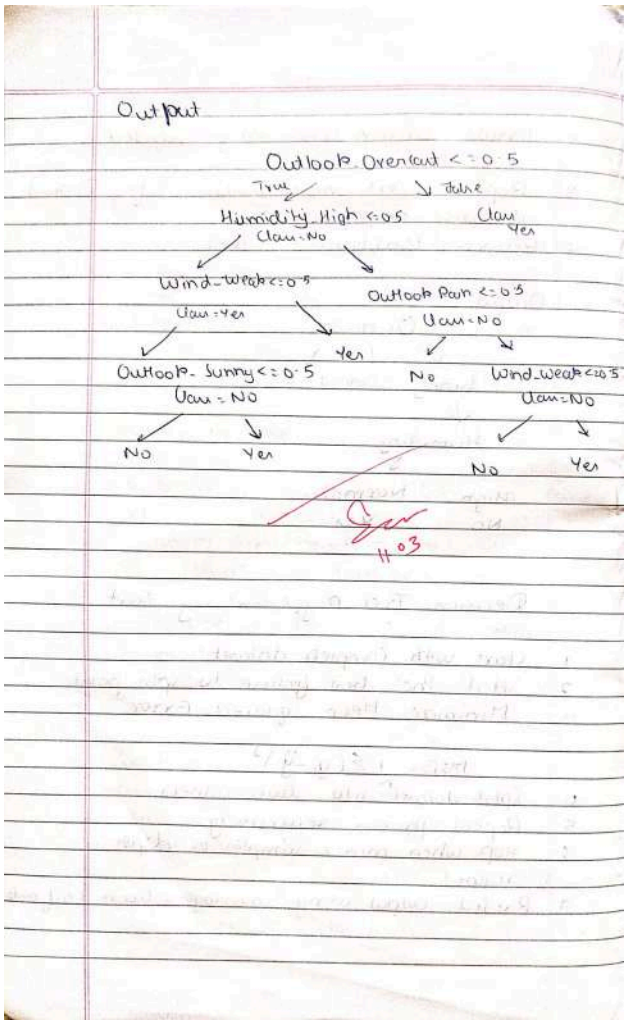
1. Start with the full Dataset D as root node
2. Check if all examples belong to same class
   * If yes, create a leaf node with that class label and stop
3. If attribute set A is empty
   * Return leafnode with majority class in D
4. Compute the entropy of Dataset D
   $E = \sum_{i=1}^{n} -p_i * \log_2(p_i)$
5. Split dataset D based on A
   Compute entropy for each subset
   Calculate Information gain
   $gain = Entropy(S) - \sum \frac{S_v}{S} Entropy(S_v)$
6. Select attribute with highest Information gain
7. Create decision tree using selected attribute
8. Repeat until no attributes left, Dataset becomes empty
9. Return Constructed tree

Output

Outlook
- Sunny → Humidity
  - High → No
  - Normal → Yes
- Overcast
- Rain

Decision Tree Regression using Cart

1. Start with complete dataset
2. Select the best feature to split point
3. Minimize Mean squared Error
   $MSE = \frac{1}{n} \sum (y_i - \hat{y})^2$
4. Split dataset into two subsets
5. Repeat process recursively
6. Stop when min samples or depth is reached
7. Predict output using average value in leaf node



Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.preprocessing import LabelEncoder
def build_classification_model(file_path, target_column):
df = pd.read_csv(file_path)
le = LabelEncoder()
for col in df.columns:
X = df.drop(target_column, axis=1)
y = df[target_column]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
classifier = DecisionTreeClassifier(criterion='entropy')
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
print(f"--- Results for {file_path} ---")
print(f"Accuracy Score: {accuracy_score(y_test, y_pred):.4f}")
print("Confusion Matrix:")
```

```
print(confusion_matrix(y_test, y_pred))
print("\n")
try:
build_classification_model('iris.csv', 'species')
build_classification_model('drug.csv', 'Drug')
except FileNotFoundError as e:
print(f"Error: {e}")
import pandas as pd
import numpy as np
3
8
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error
try:
petrol_df = pd.read_csv('petrol_consumption.csv')
X = petrol_df.drop('Petrol_Consumption', axis=1)
y = petrol_df['Petrol_Consumption']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
regressor = DecisionTreeRegressor()
regressor.fit(X_train, y_train)
y_pred = regressor.predict(X_test)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
print("--- Results for Petrol Consumption Regression ---")
print(f"Mean Absolute Error (MAE): {mae:.4f}")
print(f"Mean Squared Error (MSE): {mse:.4f}")
print(f"Root Mean Squared Error (RMSE): {rmse:.4f}")
except FileNotFoundError:
print("Error: petrol_consumption.csv not found.")
```

Program 6

Build KNN Classification model for a given dataset.

Screenshot

25/2/26

KNN

Algo:

1. Choose the number of neighbors (K)
2. Calculate the distance between new data point and the training points

$$d = \sqrt{\sum_{i=1}^{n} (x_i - x_2)^2}$$

3. Select the K nearest neighbors
4. For:
   Classification → Take the majority value
   Regression → Take the average of values

Multilinear Regression

* One dependent variable
* Multiple Independent variable

$$y = b_0 + b_1 x_1 + b_2 x_2 + \dots + b_n x_n$$

$$\theta = (X^T X)^{-1} X^T Y$$

Algorithm

1. Add bias column (1s)
2. Apply Normal Equation
3. Store Coefficients
4. Multiply to predict

Output

Traing Data

	x_1	x_2	Class
A	1	1	0
B	2	2	0
C	3	3	0
D	6	5	1
E	7	7	1
F	8	6	1

Test point (4,4)

Nearest class = C, D, B (0, 1, 0)

predicted = 0 class

Multilinear

Training Data:

Area x_1	Bedroom x_2	Price Y
1000	2	200
1200	2	250
1500	3	300
1800	3	350
2000	4	400

Predict point:- (1600, 3)

Predicted price: ~~218~~ 319.15

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import confusion_matrix, classification_report, accuracy_score
iris_df = pd.read_csv('iris.csv')
X = iris_df.drop('species', axis=1)
y = iris_df['species']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

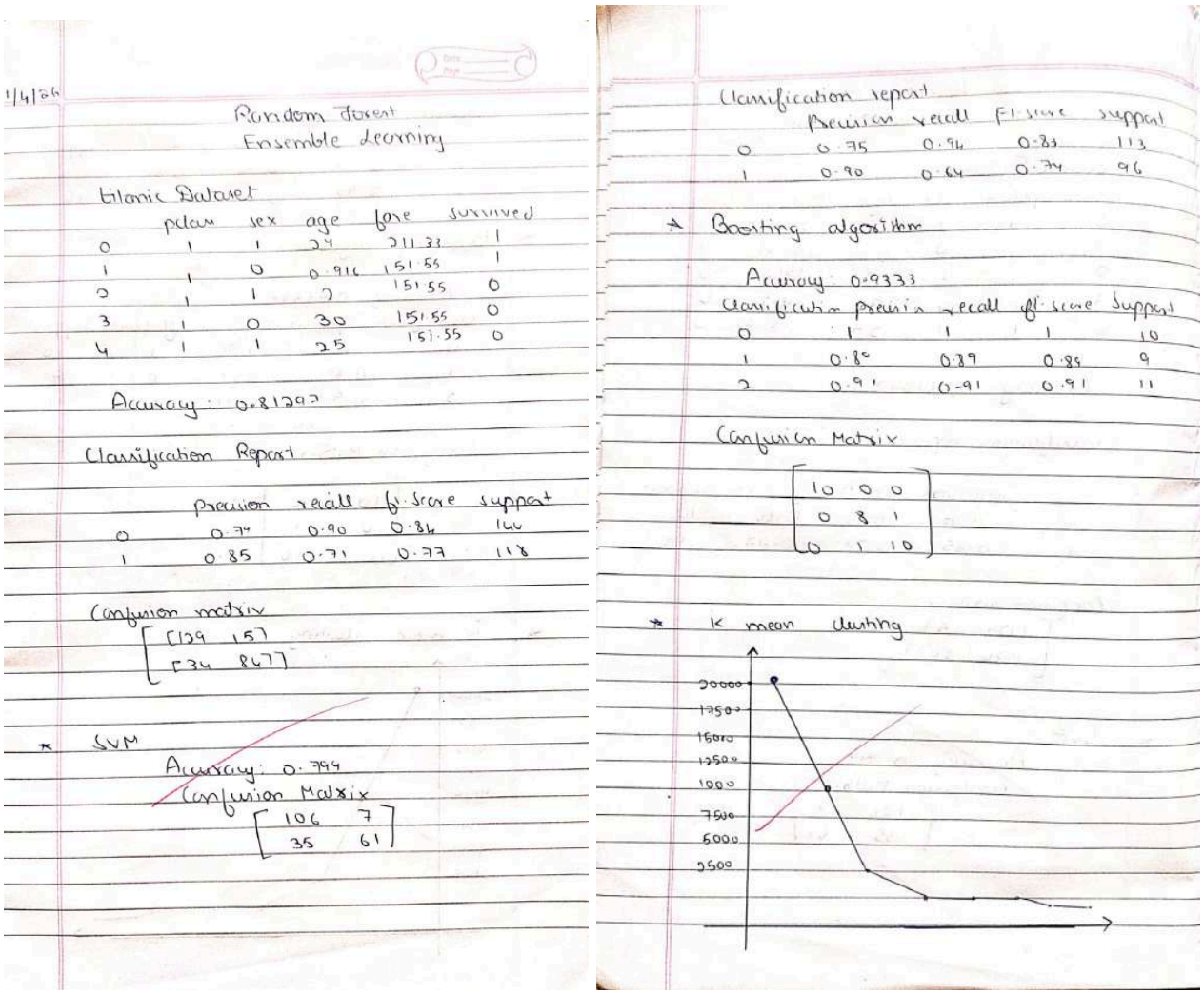
```
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)
y_pred = knn.predict(X_test)
print("--- IRIS Dataset Results ---")
print(f"Accuracy Score: {accuracy_score(y_test, y_pred)}")
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
diabetes_df = pd.read_csv('diabetes.csv')
cols_fix = ['Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI']
for col in cols_fix:
diabetes_df[col] = diabetes_df[col].replace(0, diabetes_df[col].median())
X_d = diabetes_df.drop('Outcome', axis=1)
4
2
y_d = diabetes_df['Outcome']
X_train_d, X_test_d, y_train_d, y_test_d = train_test_split(X_d, y_d, test_size=0.2,
random_state=42)
scaler_d = StandardScaler()
X_train_d = scaler_d.fit_transform(X_train_d)
X_test_d = scaler_d.transform(X_test_d)
knn_d = KNeighborsClassifier(n_neighbors=11) #
knn_d.fit(X_train_d, y_train_d)
y_pred_d = knn_d.predict(X_test_d)
print("\n--- Diabetes Dataset Results ---")
print(f"Accuracy Score: {accuracy_score(y_test_d, y_pred_d)}")
print("Confusion Matrix:\n", confusion_matrix(y_test_d, y_pred_d))
import matplotlib.pyplot as plt
import seaborn as sns
heart_df = pd.read_csv('heart.csv')
X_h = heart_df.drop('target', axis=1)
y_h = heart_df['target']
X_train_h, X_test_h, y_train_h, y_test_h = train_test_split(X_h, y_h, test_size=0.2,
random_state=42)
scaler_h = StandardScaler()
X_train_h = scaler_h.fit_transform(X_train_h)
X_test_h = scaler_h.transform(X_test_h)
scores = []
for k in range(1, 21):
knn_h = KNeighborsClassifier(n_neighbors=k)
knn_h.fit(X_train_h, y_train_h)
scores.append(knn_h.score(X_test_h, y_test_h))
4
3
best_k = scores.index(max(scores)) + 1
print(f"\nBest K value found: {best_k} with score: {max(scores)}")
final_knn = KNeighborsClassifier(n_neighbors=best_k)
final_knn.fit(X_train_h, y_train_h)
y_pred_h = final_knn.predict(X_test_h)
```

```
plt.figure(figsize=(8, 6))
sns.heatmap(confusion_matrix(y_test_h, y_pred_h), annot=True, fmt='d', cmap='Blues')
plt.title(f'Confusion Matrix (K={best_k})')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
report = classification_report(y_test_h, y_pred_h, output_dict=True)
sns.heatmap(pd.DataFrame(report).iloc[:-1, :].T, annot=True, cmap='YlGnBu')
plt.title('Classification Report')
plt.show()
```

Program 7

Build Support vector machine model for a given dataset.

Screenshot



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Random Forest
Ensemble Learning

titanic Dataset

	pclass	sex	age	fare	survived
0	1	1	24	211.33	1
1	1	0	0.916	151.55	1
2	1	1	2	151.55	0
3	1	0	30	151.55	0
4	1	1	25	151.55	0

Accuracy: 0.81297

Classification Report

	precision	recall	f1-score	support
0	0.74	0.90	0.84	146
1	0.85	0.71	0.77	118

Confusion matrix

[[129 15]
[34 84]]

* SVM

Accuracy: 0.794

Confusion Matrix

[106 7
35 61]

Classification report

	precision	recall	F1-score	support
0	0.75	0.94	0.83	113
1	0.90	0.64	0.74	96

* Boosting algorithm

Accuracy: 0.9333

Classification	precision	recall	f1-score	Support
0	1	1	1	10
1	0.80	0.89	0.85	9
2	0.91	0.91	0.91	11

Confusion Matrix

[10 0 0
0 8 1
0 1 10]

* K mean clustering

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix
df = pd.read_csv('iris.csv')
X = df.drop('species', axis=1)
y = df['species']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
svm_linear = SVC(kernel='linear')
svm_linear.fit(X_train, y_train)
y_pred_linear = svm_linear.predict(X_test)
print("--- Linear Kernel Results ---")
```

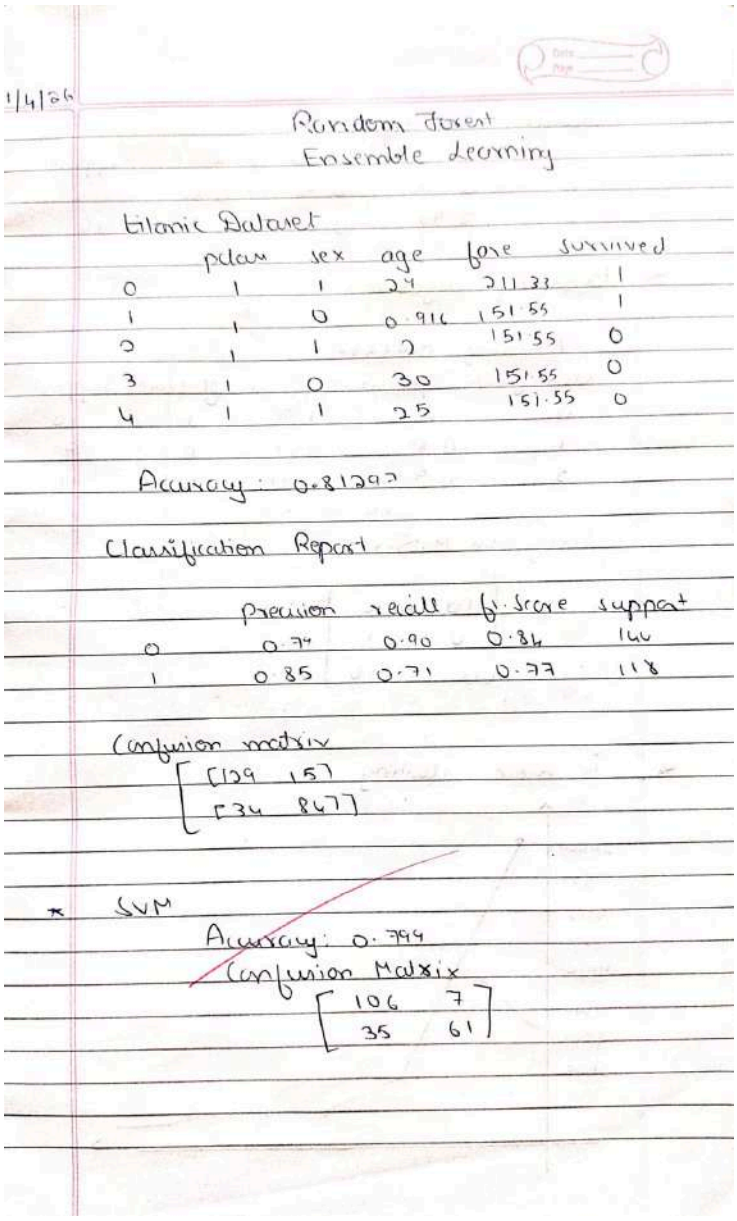
```
print(f"Accuracy Score: {accuracy_score(y_test, y_pred_linear)}")
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_linear))
svm_rbf = SVC(kernel='rbf')
svm_rbf.fit(X_train, y_train)
y_pred_rbf = svm_rbf.predict(X_test)
4
9
print("\n--- RBF Kernel Results ---")
print(f"Accuracy Score: {accuracy_score(y_test, y_pred_rbf)}")
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_rbf))
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix, roc_curve, auc
from sklearn.preprocessing import label_binarize, LabelEncoder
df_letter = pd.read_csv('letter-recognition.csv')
X_l = df_letter.drop('letter', axis=1)
y_l = df_letter['letter']
le = LabelEncoder()
y_enc = le.fit_transform(y_l)
n_classes = len(le.classes_)
X_train_l, X_test_l, y_train_l, y_test_l = train_test_split(X_l, y_enc, test_size=0.2,
random_state=42)
5
0
# 3. Build SVM Classifier (using probability=True for ROC/AUC)
svm_model = SVC(probability=True, random_state=42)
svm_model.fit(X_train_l, y_train_l)
# 4. Predictions and Scores
y_pred_l = svm_model.predict(X_test_l)
y_score_l = svm_model.predict_proba(X_test_l)
print(f"Accuracy Score: {accuracy_score(y_test_l, y_pred_l)}")
print("Confusion Matrix (Partial view):")
print(confusion_matrix(y_test_l, y_pred_l))
# 5. ROC Curve and AUC Score (Micro-average for Multiclass)
y_test_bin = label_binarize(y_test_l, classes=np.arange(n_classes))
fpr, tpr, _ = roc_curve(y_test_bin.ravel(), y_score_l.ravel())
roc_auc = auc(fpr, tpr)
print(f"Micro-average AUC Score: {roc_auc}")
# 6. Plotting
plt.figure()
plt.plot(fpr, tpr, color='darkorange', lw=2, label=f'ROC curve (area = {roc_auc:.4f})')
plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--')
5
```

```
1
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve: Letter Recognition')
plt.legend(loc="lower right")
plt.show()
```

Program 8

Implement Random forest ensemble method on a given dataset.

Screenshot



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Random Forest
Ensemble Learning

titanic Dataset

	pclass	sex	age	fare	survived
0	1	1	24	211.33	1
1	1	0	0.916	151.55	1
2	1	1	2	151.55	0
3	1	0	30	151.55	0
4	1	1	25	151.55	0

Accuracy: 0.81297

Classification Report

	precision	recall	f1-score	support
0	0.79	0.90	0.84	146
1	0.85	0.71	0.77	118

Confusion matrix

[129 15]
[34 84]]

* SVM

Accuracy: 0.799

Confusion Matrix

[106 7]
[35 61]

Code:

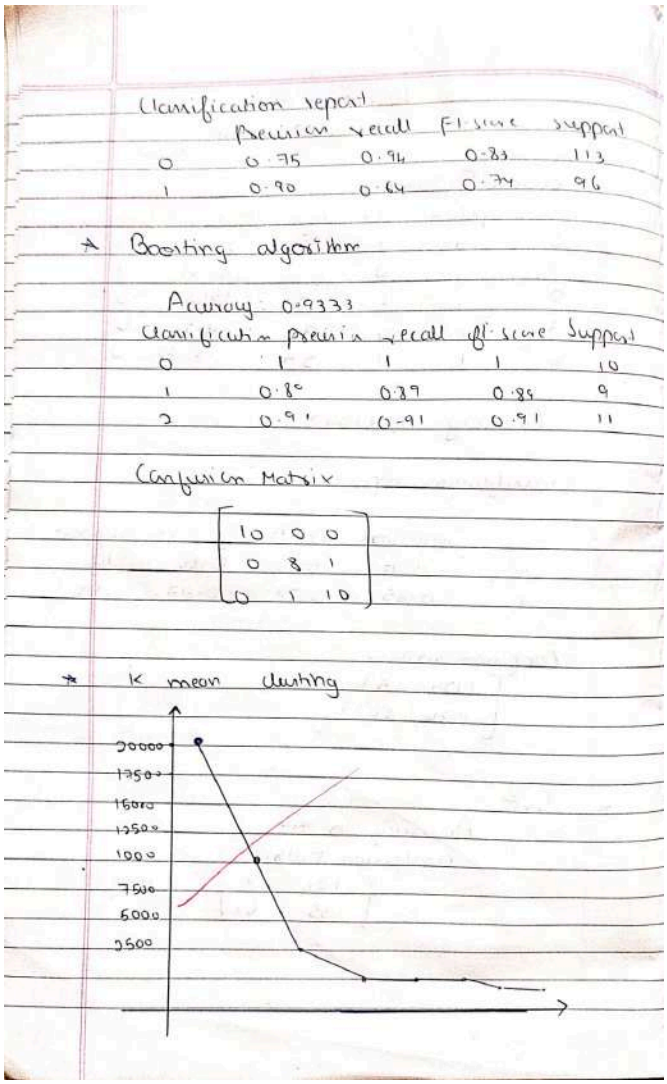
```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
df = pd.read_csv('iris.csv')
X = df.drop('species', axis=1)
y = df['species']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
rf_default = RandomForestClassifier(n_estimators=10, random_state=42)
rf_default.fit(X_train, y_train)
y_pred_default = rf_default.predict(X_test)
score_10 = accuracy_score(y_test, y_pred_default)
print(f"Accuracy with n_estimators=10: {score_10 * 100:.2f}%")
best_score = 0
best_n = 0
for n in range(1, 101):
5
6
rf = RandomForestClassifier(n_estimators=n, random_state=42)
rf.fit(X_train, y_train)
score = rf.score(X_test, y_test)
if score > best_score:
best_score = score
best_n = n
print(f"Best Accuracy Score: {best_score * 100:.2f}%")
print(f"Number of trees for best score: {best_n}")
```

Program 9

Implement Boosting ensemble method on a given dataset.

Screenshot



Classification report

	Precision	recall	F1 score	support
0	0.75	0.94	0.83	113
1	0.90	0.64	0.74	96

* Boosting algorithm

Accuracy 0.9333

Classification	precision	recall	f1 score	Support
0	1	1	1	10
1	0.80	0.89	0.85	9
2	0.91	0.91	0.91	11

Confusion Matrix

$$\begin{bmatrix} 10 & 0 & 0 \\ 0 & 8 & 1 \\ 0 & 1 & 10 \end{bmatrix}$$

* K mean clustering

Code:

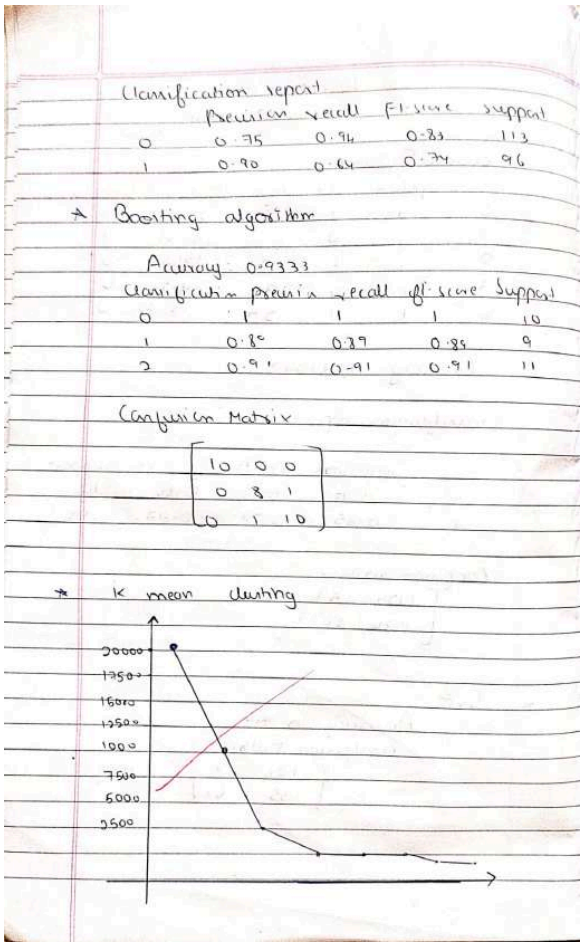
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import accuracy_score
df = pd.read_csv('income.csv')
# Separate features (X) and target (y)
X = df.drop('income_level', axis=1)
y = df['income_level']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
ada_10 = AdaBoostClassifier(n_estimators=10, random_state=42)
```

```
ada_10.fit(X_train, y_train)
y_pred_10 = ada_10.predict(X_test)
score_10 = accuracy_score(y_test, y_pred_10)
print(f"Prediction score with 10 trees: {score_10:.4f}")
# Testing various numbers of trees to identify the best performance
tree_counts = [5, 10, 25, 50, 100, 150, 200, 300, 400, 500]
scores = []
print("\nFine-tuning results:")
for n in tree_counts:
model = AdaBoostClassifier(n_estimators=n, random_state=42)
model.fit(X_train, y_train)
acc = model.score(X_test, y_test)
scores.append(acc)
print(f"Trees: {n:3} | Accuracy: {acc:.4f}")
best_acc = max(scores)
best_n = tree_counts[scores.index(best_acc)]
6
1
print(f"\nOptimal performance: {best_acc:.4f} accuracy achieved with {best_n} trees.")
# 5. Visualization (Optional but helpful)
plt.figure(figsize=(8, 5))
plt.plot(tree_counts, scores, marker='o', linestyle='-', color='darkgreen')
plt.title('AdaBoost Accuracy vs. Number of Trees')
plt.xlabel('n_estimators')
plt.ylabel('Test Accuracy')
plt.grid(True)
plt.show()
```

Program 10

Build k-Means algorithm to cluster a set of data stored in a .CSV file.

Screenshot



Classification report

	Precision	recall	F1 score	support
0	0.75	0.94	0.83	113
1	0.90	0.64	0.74	96

* Boosting algorithm

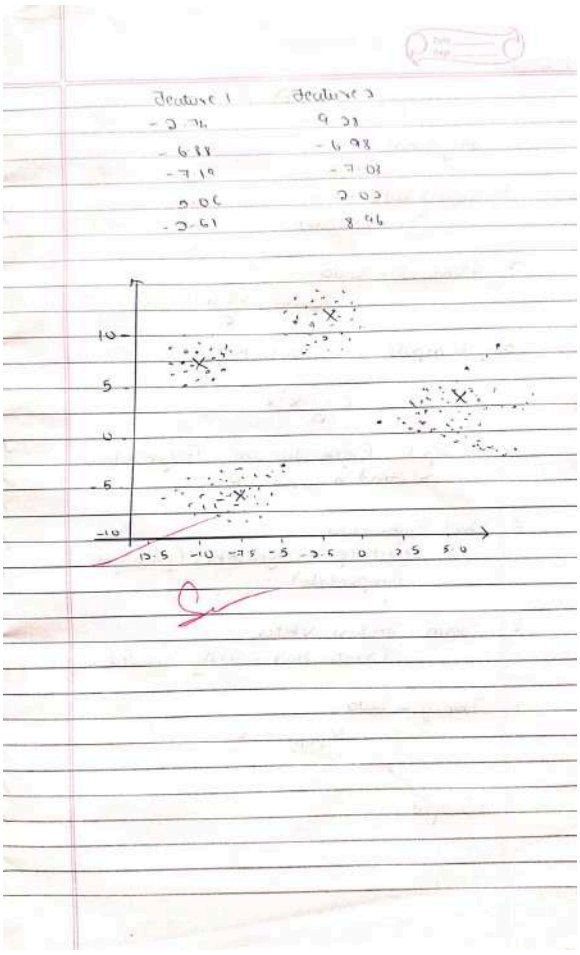
Accuracy 0.9333

Classification	precision	recall	f1 score	Support
0	1	1	1	10
1	0.80	0.89	0.85	9
2	0.91	0.91	0.91	11

Confusion Matrix

$$\begin{bmatrix} 10 & 0 & 0 \\ 0 & 8 & 1 \\ 0 & 1 & 10 \end{bmatrix}$$

* K mean clustering



Feature 1	Feature 2
-2.74	9.23
-6.88	-6.98
-7.19	-7.03
5.06	2.03
-2.61	8.46

Code:

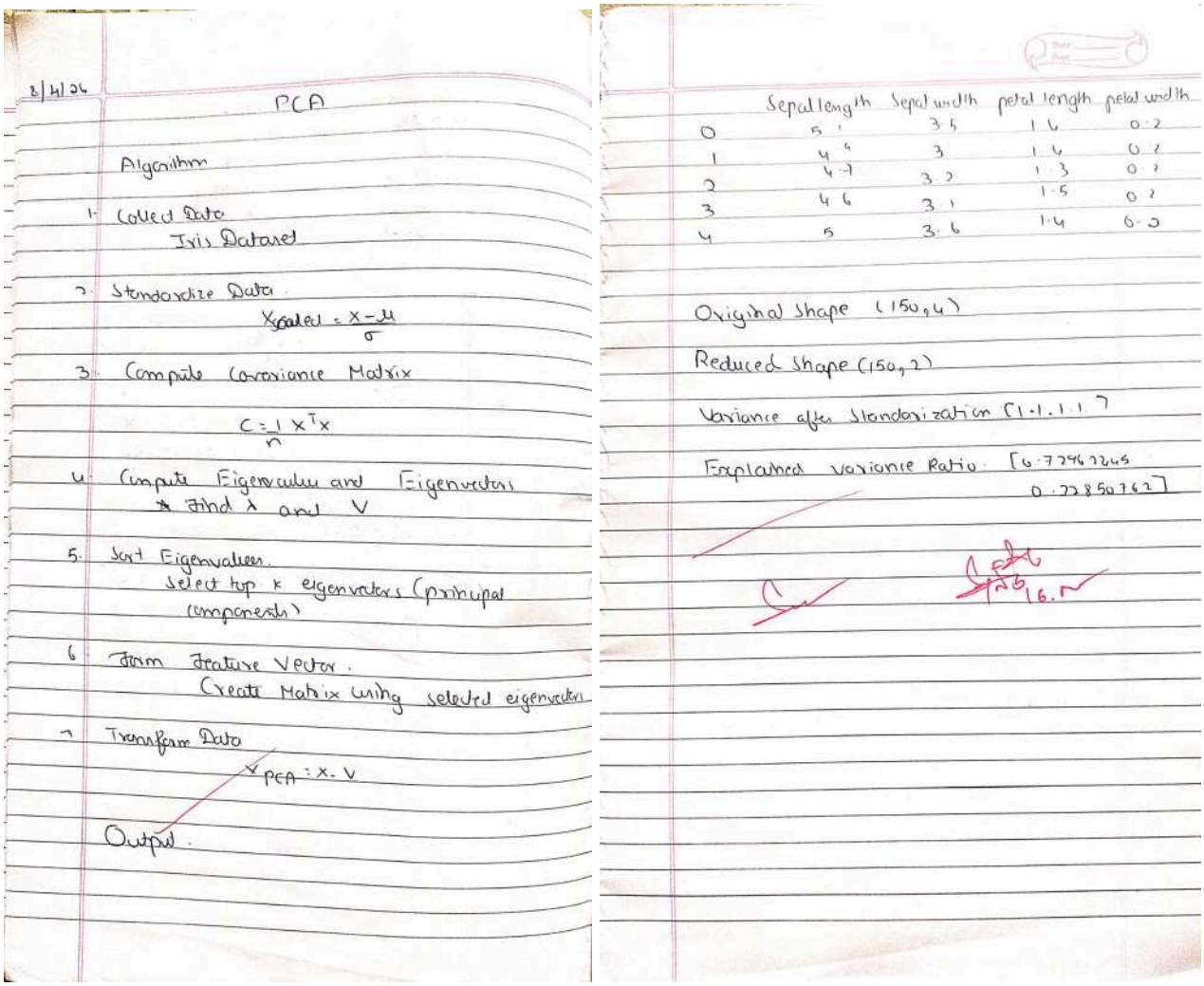
```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
from sklearn.decomposition import PCA
df = pd.read_csv('heart.csv')
categorical_features = ['cp', 'restecg', 'slope', 'ca', 'thal']
df_encoded = pd.get_dummies(df, columns=categorical_features, drop_first=True)
X = df_encoded.drop('target', axis=1)
y = df_encoded['target']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
6
7
models = {
"Logistic Regression": LogisticRegression(random_state=42),
"SVM": SVC(random_state=42),
"Random Forest": RandomForestClassifier(random_state=42)
}
print("--- Accuracies Before PCA ---")
results_before = {}
for name, model in models.items():
model.fit(X_train_scaled, y_train)
y_pred = model.predict(X_test_scaled)
acc = accuracy_score(y_test, y_pred)
results_before[name] = acc
print(f"{name}: {acc:.4f}")
pca = PCA(n_components=0.95)
X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)
print(f"\nOriginal dimensions: {X_train_scaled.shape[1]}")
print(f"Dimensions after PCA: {X_train_pca.shape[1]}")
print("\n--- Accuracies After PCA ---")
6
8
results_after = {}
for name, model in models.items():
model.fit(X_train_pca, y_train)
y_pred = model.predict(X_test_pca)
acc = accuracy_score(y_test, y_pred)
results_after[name] = acc
print(f"{name}: {acc:.4f}")
```

Program 10

Implement Dimensionality reduction using Principal Component Analysis (PCA) method.

Screenshot



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PCA

Algorithm

1. Collect Data
   Iris Dataset
2. Standardize Data
   $X_{scaled} = \frac{X-\mu}{\sigma}$
3. Compute Covariance Matrix
   $C = \frac{1}{n} X^T X$
4. Compute Eigenvalue and Eigenvectors
   * Find λ and V
5. Sort Eigenvalues.
   Select top k eigenvectors (principal components)
6. Form Feature Vector.
   Create Matrix using selected eigenvectors
7. Transform Data
   $X_{PCA} = X \cdot V$

Output.

	Sepal length	Sepal width	petal length	petal width
0	5.1	3.5	1.6	0.2
1	4.9	3	1.6	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5	3.6	1.4	0.2

Original Shape (150,4)

Reduced Shape (150,2)

Variance after Standarization [1.1.1.1]

Explained variance Ratio: [0.72962245 0.22850762]

16.5

Code:

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
from sklearn.decomposition import PCA
df = pd.read_csv('heart.csv')
categorical_cols = ['cp', 'restecg', 'slope', 'ca', 'thal']
df_encoded = pd.get_dummies(df, columns=categorical_cols, drop_first=True)
```

```
X = df_encoded.drop('target', axis=1)
y = df_encoded['target']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
7
4
def evaluate_models(X_tr, X_te, y_tr, y_te):
models = {
"SVM": SVC(),
"Logistic Regression": LogisticRegression(),
"Random Forest": RandomForestClassifier(random_state=42)
}
accuracies = {}
for name, model in models.items():
model.fit(X_tr, y_tr)
y_pred = model.predict(X_te)
accuracies[name] = accuracy_score(y_te, y_pred)
return accuracies
results_initial = evaluate_models(X_train_scaled, X_test_scaled, y_train, y_test)
print("--- Accuracies Without PCA ---")
for model, acc in results_initial.items():
print(f"{model}: {acc:.4f}")
pca = PCA(n_components=0.95)
X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)
print(f"\nOriginal dimensions: {X_train_scaled.shape[1]}")
7
5
print(f"Dimensions after PCA: {X_train_pca.shape[1]}")
results_pca = evaluate_models(X_train_pca, X_test_pca, y_train, y_test)
print("\n--- Accuracies With PCA ---")
for model, acc in results_pca.items():
print(f"{model}: {acc:.4f}")
```